

MTA Daily Ridership — COVID-19 Recovery Analysis

Technical Report

DEPI Data Analysis Track

1. Executive Summary

COVID-19 hit New York's public transit harder than almost any event in its history. In just five weeks in early 2020, subway ridership collapsed by 96% — from 5.5 million daily riders to fewer than 200,000. Since then, recovery has been uneven and structurally significant: some modes fully rebounded while others stabilized at permanently lower baselines, reflecting deep behavioral shifts in how New Yorkers live, work, and move.

This report presents a full data analysis of MTA ridership from March 2020 to October 2024 across 7 transit modes, examining the collapse, the recovery trajectory, and the long-term implications for transit planning in a post-pandemic city.

“COVID didn't just temporarily disrupt transit — it accelerated behavioral changes in remote work, car usage, and paratransit demand that were already quietly underway.”

Headline KPI cards

Metric	Value
Subway ridership crash	5.5M → 198K (−96%), Mar→Apr 2020
Fastest recovery	93.4% avg — Bridges & Tunnels
Slowest recovery	~37.6% avg — Staten Island Railway
Exceeded pre-pandemic	126% — Access-A-Ride, 2024

2. Dataset Overview & Business Constraints

2.1 About the Dataset

The dataset is sourced from the MTA Open Data Portal and covers daily ridership for all major New York City transit modes from March 1, 2020 through October 31, 2024 — a total of 1,706 calendar days.

Column	Description	Type	Note
Date	Calendar date of observation	Date	Daily granularity
Subways: Total Estimated Ridership	Estimated daily subway riders	Integer	Core metric
Subways: % of Comparable Pre-Pandemic Day	Recovery % vs equivalent 2019 day	Integer (0–143)	Keep as int
Buses: Total Estimated Ridership	Daily bus passengers	Integer	—
Buses: % of Comparable Pre-Pandemic Day	Recovery %	Integer	Keep as int
LIRR / Metro-North / Access-A-Ride	Same structure per mode	Integer	—
Bridges and Tunnels: Total Traffic	Vehicle crossings per day	Integer	Not riders — vehicles
Staten Island Railway: Total Estimated Ridership	Smallest mode	Integer	—

Long-Format Table for Cross-Mode Visuals

The wide, per-mode columns above are kept for mode-specific calculations, but they cannot directly power visuals that compare across modes on one axis (e.g. “Sum of Total Ridership by Transport Mode,” the Transport Mode Category pie charts, or the Recovery Rate charts). For these, the data was additionally unpivoted into a long-format table with columns Date, Transport Mode, Total Ridership, and % of Pre-Pandemic Day, plus a Transport Mode Category mapping (Rapid Transit / Public Transit / Infrastructure Usage). This unpivoted table is the source for every chart on dashboard pages 2, 3, and 4.

2.2 Business Constraints & Analytical Challenges

Integer % Columns — Correct by Design

The “% of Pre-Pandemic” columns store whole numbers (e.g. 97, 143) — not decimals. This is intentional: the values represent percentage points, not ratios. Converting them to float (0.97) would break all percentage-based visuals. Decision: keep as int, apply % label in formatting only.

Ridership ≠ Vehicle Traffic

Bridges & Tunnels measures vehicle crossings, not passengers. This makes direct comparisons with other modes misleading. In Power BI, this required separate axis scaling and clear labeling to prevent misinterpretation.

Estimated Ridership — Not Exact Counts

All ridership figures are estimates based on MetroCard swipes and OMNY taps — they exclude fare evaders, paratransit no-shows, and irregular entries. This limits precision but is sufficient for trend analysis.

Day-of-Week Comparability

Pre-pandemic % is calculated by comparing each day to the same day of week in the same calendar period of 2019. This controls for weekend effects, holidays, and seasonal variation — but means a “100%” value on a quiet Sunday ≠ 100% on a busy Monday.

Data Coverage Gap — Bus Ridership April–August 2020

Bus ridership data shows near-zero values (under 6,000) for April–July 2020, then jumps to ~935,000 in September. This appears to be a reporting methodology change by MTA, not an actual ridership anomaly. These months were excluded from bus trend analysis.

No External Variables in Dataset

The dataset contains only ridership counts — no employment data, remote work rates, weather, or policy events. All behavioral interpretations (WFH effect, car preference) are inferred from patterns, not directly measurable from this data alone.

3. What We Built in Power BI

The Power BI dashboard was designed to make 1,706 days of transit data instantly navigable for a non-technical audience — giving decision-makers a clear view of recovery progress across all 7 transit modes without requiring any data background. The dashboard has 4 pages (see Section 3.3, corrected below).

3.1 Data Model & Preparation

Step 1 — Data Import: Loaded the MTA CSV directly into Power BI via Get Data → Text/CSV. The date column was automatically parsed as a Date type.

Step 2 — Calendar Table: Created a Date table using CALENDARAUTO() DAX, so the table auto-spans every year present in the data (2020–2024) without hardcoding start/end dates. Calculated columns added: Year, Month Name (FORMAT([Date], "MMMM")), Day Name (FORMAT([Date], "dddd")), and WeekEnd? (a Yes/No indicator used by the Weekend/Weekday Ridership cards and the “Post/Pre Pandemic by WeekEnd” chart on page 3).

Step 3 — Column Typing: All ridership columns kept as Whole Number. All % columns kept as Whole Number with % display format applied at the visual level — NOT converted to decimal, to avoid the ×100 multiplication error.

Step 4 — Unpivoted Table: Built a long-format table (Date, Transport Mode, Total Ridership, % of Pre-Pandemic) to support the Transport Mode, Transport Mode Category, and Season breakdowns used throughout pages 2–4. Transport Mode Category (Rapid Transit / Public Transit / Infrastructure Usage) and Season (Spring/Summer/Autumn/Winter, from the Date table's month) were added as calculated columns on this table.

Step 5 — Measures: Created DAX measures for total ridership, weekday/weekend splits, pre-/post-pandemic splits, and recovery rate — see Section 3.2.

3.2 Key DAX Measures Used

Measure / Column	DAX Formula	Used In
Total Ridership (measure)	SUM(Unpivoted[Total Ridership])	Sum of Total Ridership by Day Name; by Transport Mode (page 3)
Weekday Ridership (measure)	CALCULATE([Total Ridership], 'Date'[WeekEnd?] = "No")	Weekday Ridership KPI card (page 3)
Weekend Ridership (measure)	CALCULATE([Total Ridership], 'Date'[WeekEnd?] = "Yes")	Weekend Ridership KPI card (page 3)
Pre Pandemic (measure)	CALCULATE([Total Ridership], Unpivoted[Period] = "Pre Pandemic")	25bn KPI card, WeekEnd grouped bar, Category pie, Seasons treemap (pages 1–2)
Post Pandemic (measure)	CALCULATE([Total Ridership], Unpivoted[Period] = "Post Pandemic")	8bn KPI card, Post Pandemic by Year bar, Category pie, Seasons treemap (pages 1–2)
Recovery Rate (measure)	DIVIDE([Post Pandemic], [Pre Pandemic] + [Post Pandemic])	Recovery Rate by Transport Mode; by Transport Mode and Year (page 4)
Transport Mode Category (calculated column)	SWITCH(Unpivoted[Transport Mode], "Subways", "Rapid Transit", "LIRR", "Rapid Transit", "Metro-North", "Rapid Transit", "Buses", "Public Transit", "Access-A-Ride", "Public Transit", "Bridges and Tunnels", "Infrastructure Usage", "Staten Island Railway", "Rapid Transit")	Post/Pre Pandemic by Transport Mode Category pie charts (page 2)
Season (calculated column)	SWITCH(TRUE(), MONTH('Date'[Date]) IN {12,1,2}, "Winter", MONTH('Date'[Date]) IN {3,4,5}, "Spring", MONTH('Date'[Date]) IN {6,7,8}, "Summer", "Autumn")	Post/Pre Pandemic by Seasons treemaps (page 2)
Day Name (calculated column)	FORMAT('Date'[Date], "dddd")	Sum of Total Ridership by Day Name; Pre Pandemic by Day Name and Transport Mode (pages 1, 3)
WeekEnd? (calculated column)	IF(WEEKDAY('Date'[Date],2) >= 6, "Yes", "No")	Weekday/Weekend KPI cards; Post/Pre Pandemic by WeekEnd chart (page 3)

3.3 Dashboard Pages & Visuals

Page 1 — Pandemic Overview

- KPI Cards: 25bn Pre Pandemic total, 8bn Post Pandemic total
- Bar Chart: Post Pandemic ridership by Year (2020–2024)
- 100% Stacked Bar: Pre Pandemic ridership by Day Name and Transport Mode

Page 2 — Category & Seasonality

- Pie Chart: Post Pandemic ridership by Transport Mode Category (Rapid Transit 59.46%, Public Transit 22.13%, Infrastructure Usage 18.41%)
- Pie Chart: Pre Pandemic ridership by Transport Mode Category (Rapid Transit 69.52%, Public Transit 18.73%, Infrastructure Usage 11.75%)
- Treemap: Post Pandemic ridership by Season
- Treemap: Pre Pandemic ridership by Season
- Year range slicer (2020–2024) filtering the whole page

Page 3 — Ridership Patterns

- KPI Cards: Weekend Ridership (1.61bn), Weekday Ridership (6.33bn)
- Bar Chart: Sum of Total Ridership by Day Name
- Grouped Bar: Post Pandemic and Pre Pandemic ridership by WeekEnd? (Yes/No)

- Bar Chart: Sum of Total Ridership by Transport Mode (Subways highest, then Buses, Bridges and Tunnels, LIRR, Metro-North, Access-A-Ride, Staten Island Railway)

Page 4 — Recovery Rate

- Area Chart: Recovery Rate by Transport Mode (Bridges and Tunnels highest, Staten Island Railway lowest)
- Multi-line Chart: Recovery Rate by Transport Mode and Year, one line per year 2020–2024

4. Key Findings & Insights

The data analysis produced 4 major findings, each supported by specific patterns in the ridership dataset:

Finding 1: Cars Won the Pandemic

Bridges & Tunnels — representing personal vehicle traffic — recovered to 93.4% average across the full period, while subways averaged only 55.5%. The gap in 2020 alone was 42 percentage points (68% vs 26%). When given a choice, New Yorkers consistently chose driving over shared transit. By 2024, B&T exceeded 100% of pre-pandemic levels while subways remained at 71%.

Finding 2: Remote Work Created a New Permanent Baseline

Metro-North — which carries suburban commuters — averaged only 51.1% of pre-pandemic levels across the full period. This is not a slow recovery: it is a new structural equilibrium. Hybrid work schedules (2–3 days in office) have become permanent for suburban professionals. The data shows Metro-North stopped recovering around 2023 and stabilized, suggesting this IS the new normal.

Finding 3: Paratransit Demand Surged Beyond Pre-Pandemic

Access-A-Ride — serving mobility-limited riders — is the only mode that exceeded 100% of pre-pandemic levels, reaching 126% by 2024. This reflects three factors: NYC's aging population, expanded eligibility criteria post-COVID, and pent-up demand from the period when services were restricted. This is a planning challenge, not just good news.

Finding 4: Staten Island Railway Marks the Recovery Floor

Subway ridership recovered steadily from 2020 to 2023, then plateaued around 70–71% of pre-pandemic levels — tourism and essential-worker trips came back, but daily white-collar office commutes did not fully return. Looking across all modes, Staten Island Railway sits at the very bottom of the recovery curve (~37.6% average, ~50% by 2024), making it the slowest-recovering mode overall, with Metro-North close behind.

4.1 Recovery Scorecard by Mode

Transit Mode	2020	2021	2022	2023	2024	Status
Bridges & Tunnels	68%	87%	97%	100%	~101%	Fully Recovered
Access-A-Ride	55%	72%	87%	98%	~126%	Exceeded Baseline
LIRR	22%	43%	64%	72%	~74%	Strong Recovery
Subways	26%	46%	61%	70%	~71%	Partial Recovery
Buses	29%	48%	60%	69%	~69%	Partial Recovery
Metro-North	18%	38%	57%	65%	~67%	Stabilized (Low)
Staten Island Railway	19%	31%	41%	48%	~50%	Significantly Below — slowest overall

5. Conclusions & Strategic Implications

The MTA isn't just recovering from COVID — it is adapting to a post-COVID city that looks structurally different from the one that existed in February 2020. Three things stand out:

The Office Commute May Never Fully Return

Metro-North and LIRR data shows clearly that the Monday-to-Friday, 9-to-5 commute pattern has changed permanently. Hybrid work is not a temporary phase — it is the new contract between employers and suburban workers. Transit planning needs to model for a 3-day commute week, not 5.

Local & Essential Trips Are the New Backbone

Subway and bus ridership at ~70% is being driven by essential workers, students, and local trips — not white-collar office commuters. This is a more diverse and potentially more recession-resistant ridership base than the old commuter-heavy model. The MTA should invest in reliability and off-peak service for this audience.

Paratransit Is a Growing Obligation

Access-A-Ride exceeding 100% capacity is both good news and a planning challenge. As NYC's population ages and eligibility expands, paratransit demand will keep growing. Proactive investment in this service is not optional — it is a civil rights and mobility equity obligation.

6. Data Notes & Methodology

This analysis uses the MTA's publicly available Daily Ridership dataset from the MTA Open Data Portal. Recovery percentages compare each day's ridership to the equivalent pre-pandemic day (same day of week, same seasonal period, baseline: early 2020 before COVID restrictions). An average of 100% indicates full recovery; above 100% means demand exceeded the pre-pandemic baseline. Bus ridership anomalies in April–August 2020 are attributed to a reporting methodology change by MTA and should be interpreted with caution.

Tools used: Python (pandas, reportlab, openpyxl) for data cleaning and file generation; Power BI for interactive dashboards and DAX measures (including CALENDARAUTO() date table and an unpivoted Transport Mode table); Excel for pivot table analysis; HTML/JavaScript for web-based dashboard delivery.